

UNIVERSIDAD DE GUADALAJARA



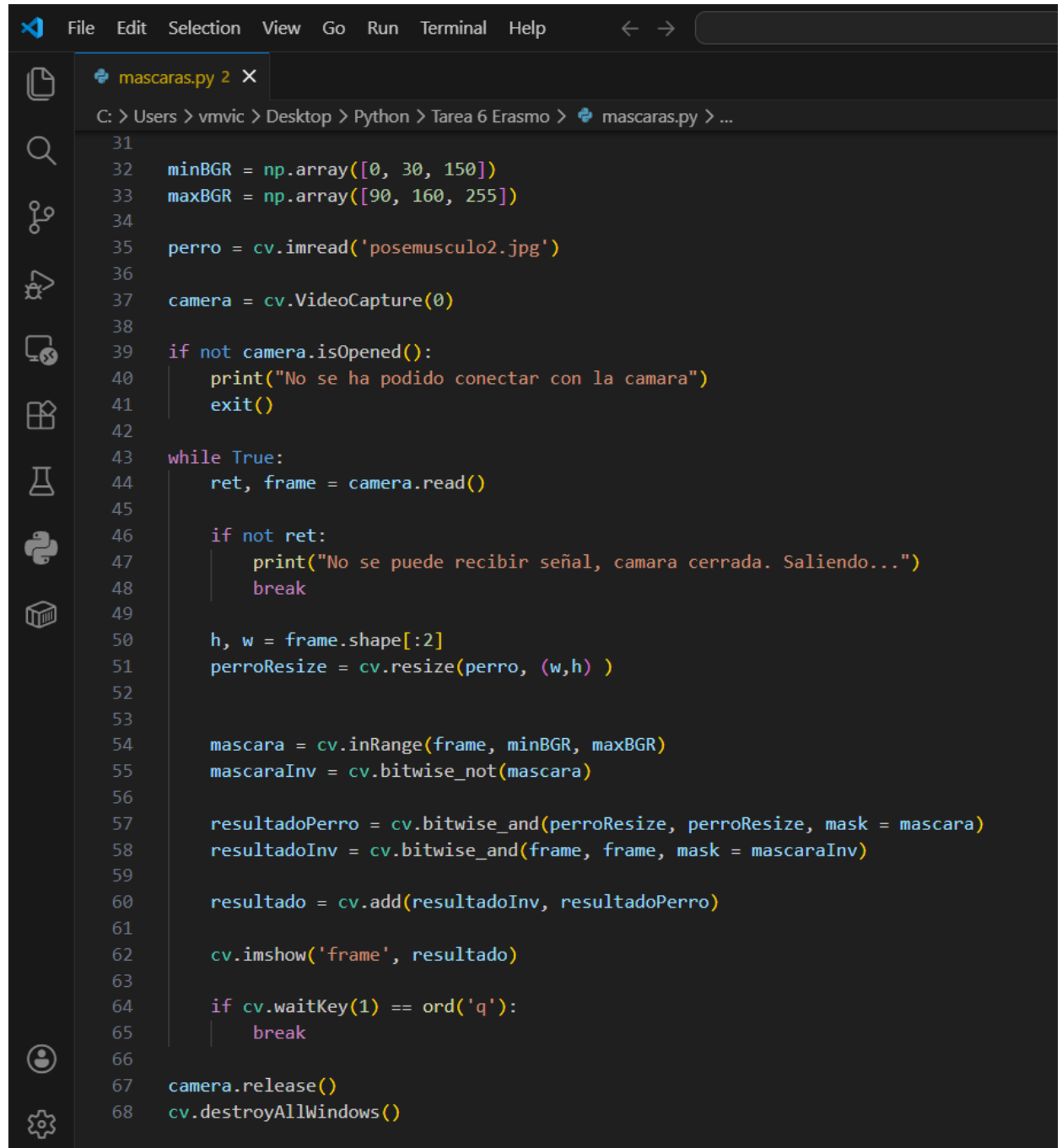
CENTRO UNIVERSITARIO DE CIENCIAS EXACTAS E INGENIERÍAS

Análisis y algoritmos

Reporte de práctica

Nombre del alumno:	Victor Manuel Quijas Contreras
Código del alumno:	220429771
Profesor:	Erasmus Gabriel Martínez Soltero
Título de la práctica:	"Video cambiando el fondo"
Fecha:	12 de Marzo 2026

Código

A screenshot of a code editor window with a dark theme. The window has a menu bar at the top with options: File, Edit, Selection, View, Go, Run, Terminal, and Help. Below the menu bar is a tab labeled 'mascaras.py 2 x'. The main area shows a Python script with line numbers from 31 to 68. The script uses OpenCV (cv) and NumPy (np) to process an image of a dog. It sets color ranges for a mask, reads an image, captures video from a camera, and performs bitwise operations to create a mask and combine it with the original image. The script includes error handling for camera connection and signal reception. The left sidebar contains icons for file explorer, search, source control, and other IDE features.

```
31
32 minBGR = np.array([0, 30, 150])
33 maxBGR = np.array([90, 160, 255])
34
35 perro = cv.imread('posemusculo2.jpg')
36
37 camera = cv.VideoCapture(0)
38
39 if not camera.isOpened():
40     print("No se ha podido conectar con la camara")
41     exit()
42
43 while True:
44     ret, frame = camera.read()
45
46     if not ret:
47         print("No se puede recibir señal, camara cerrada. Saliendo...")
48         break
49
50     h, w = frame.shape[:2]
51     perroResize = cv.resize(perro, (w,h) )
52
53
54     mascara = cv.inRange(frame, minBGR, maxBGR)
55     mascaraInv = cv.bitwise_not(mascara)
56
57     resultadoPerro = cv.bitwise_and(perroResize, perroResize, mask = mascara)
58     resultadoInv = cv.bitwise_and(frame, frame, mask = mascaraInv)
59
60     resultado = cv.add(resultadoInv, resultadoPerro)
61
62     cv.imshow('frame', resultado)
63
64     if cv.waitKey(1) == ord('q'):
65         break
66
67 camera.release()
68 cv.destroyAllWindows()
```

Figura 1: Código

Link del video

<https://www.youtube.com/watch?v=yqffAKXd9Z4>